Records and Unit Values of an XOR-Triangle Boundary

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

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Abstract

We prove both conjectures in OEIS A334595: its record positions are one and $2^m + 1$ for $m \geq 1$, and its value equals one exactly at powers of two. The right boundary of an XOR triangle is a binomial transform over the field with two elements. Its inverse is the same transform, which determines every value, every fixed-length image, and the unique maximum of each length block.

1 The boundary transform

For $n \geq 1$, form the XOR triangle whose first row is the ordinary binary expansion of n. Each entry in a subsequent row is the sum modulo two of the two entries above it. Read the right boundary from the upper right corner to the bottom as a binary word; its integer value is $a(n)$ in OEIS A334595.

Write $n = \sum_{i=0}^{L-1} b_i 2^i$, with $b_{L-1} = 1$, and let y_j be the rightmost entry in row j, starting with $j = 0$. Repeated addition of neighboring entries gives

$$y_j = \sum_{i=0}^{j} \binom{j}{i} b_i \pmod 2, \qquad a(n) = \sum_{j=0}^{L-1} y_j 2^{L-1-j}. \tag{1}$$

Pascal's identity proves the first expression inductively. This is a finite explicit all-index formula, independent of constructing the interior of the triangle.

Lemma 1. *The map $T : (b_i) \mapsto (y_j)$ in (1) is its own inverse on every fixed length L over $\mathbf{F}_2$.*

Proof. For $0 \leq i \leq j < L$, the coefficient of b_i in $(T^2 b)_j$ is

$$\sum_{k=i}^{j} \binom{j}{k}\binom{k}{i} = \binom{j}{i} \sum_{k=i}^{j} \binom{j-i}{k-i} = \binom{j}{i} 2^{j-i}.$$

Modulo two this equals one for $j = i$ and zero for $j > i$. □

2 All values equal to one

Theorem 1. *For every positive integer n, $a(n) = 1$ if and only if n is a power of two.*

Proof. On length L, output one has boundary vector $(y_0, \ldots, y_{L-1}) = (0, \ldots, 0, 1)$. Its inverse under T is that same vector: for $j < L - 1$ every input summand is zero, and at $j = L - 1$ the diagonal entry is one. Thus $b_{L-1} = 1$ is its only nonzero input bit, giving $n = 2^{L-1}$. Conversely this input clearly gives output one. □

3 Every record position and value

Theorem 2. *For every $L \geq 2$, the maximum of $a(n)$ on $2^{L-1} \leq n < 2^L$ is $2^L - 2$, attained uniquely at $n = 2^{L-1} + 1$. Therefore the strict record positions of A334595 are exactly*

$$1, \quad 2^m + 1 \quad (m \geq 1),$$

with respective record values 1 and $2^{m+1} - 2$.

Proof. All outputs are represented by L bits, so at most $2^L - 1$. The all-one boundary vector has inverse with

$$b_0 = 1, \qquad b_j = \sum_{i=0}^{j} \binom{j}{i} = 2^j = 0 \pmod 2 \quad (j \geq 1).$$

For $L \geq 2$ its leading input bit is zero; hence output $2^L - 1$ is impossible on this input length. The next possible output, $2^L - 2$, has vector $(1, \ldots, 1, 0)$. It differs from the all-one vector only in its last coordinate, so its inverse has exactly the first and last coordinates equal to one. This is the valid input $n = 2^{L-1} + 1$, unique by Lemma 1. It is therefore the unique block maximum.

The first input in each block is 2^{L-1} and has output one by the preceding theorem. The second input already attains the block maximum, which exceeds the maxima of all previous length blocks. No later input in that block can set a new strict record. The initial block consists of $n = 1$ with output one. □

The fixed-length image itself also has an exact description: it consists of the L-bit words y, allowing leading zeros, that satisfy

$$\sum_{i=0}^{L-1} \binom{L-1}{i} y_i = 1 \pmod 2.$$

This is precisely the condition that their unique inverse has leading bit one. Thus no exceptional word or index is hidden in the characterization above.

4 Source and independent checks

The entry's linked MathOverflow discussion concerns fixed points of a related boundary rotation. Gerry Myerson's May 2, 2020 comment observes the binomial nature of XOR differences. We use that standard transform explicitly and prove the two specific conjectures still stated in A334595; no priority for the transform itself is claimed. The verifier compares all 8191 official terms with direct triangle construction and an independent binomial-coefficient implementation. It checks both conjectures and the unique block maxima through $n = 65535$, and verifies $T^2 = I$ on every binary vector of lengths one through ten. The all-index conclusions follow from the proofs.

References

[1] The OEIS Foundation Inc., A334595 and its 8191-term b-file, accessed October 26, 2026.

[2] DSM and contributors, *Number triangle*, https://mathoverflow.net/questions/359138/number-triangle, May 2020, accessed October 26, 2026.